PHD115 Fundamentals of Chemistry · Chemical Reactions I

After class: check your understanding

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 30 minutes. Questions follow the seven parts of the lecture and end with a pharmacy scenario.

Answer without notes first, then use the slides to check. Use the atomic masses given in each question. A periodic table and the answer key are at the back.

Parts 1–2 · Equations and ratios

Q1 CR1-LO1

Which is the balanced equation for iron rusting, $Fe + O_2 \rightarrow Fe_2O_3$?

A. $2Fe + O_2 \rightarrow Fe_2O_3$

B. $4Fe + 3O_2 \rightarrow 2Fe_2O_3$

C. $Fe + O_3 \rightarrow FeO_3$

D. $2Fe + 3O_2 \rightarrow Fe_2O_3$

Q2 CR1-LO1

Balance $N_2 + __H_2 \rightarrow 2NH_3$. What coefficient goes in front of H_2?

Q3 CR1-LO1

It is acceptable to change a subscript in a formula to balance an equation.

True / False

Q4 CR1-LO1

12.0 g of magnesium burns completely in 8.0 g of oxygen to form magnesium oxide only. What mass of magnesium oxide forms?

Q5 CR1-LO2

Using $2C_2H_2 + 5O_2 \rightarrow 4CO_2 + 2H_2O$, how many O_2 molecules react with 10 C_2H_2 molecules?

Part 3 · Types of reaction

Q6 CR1-LO3

Classify each reaction.

Options: combination · decomposition · combustion · single displacement · double displacement

$2H_2O_2 \rightarrow 2H_2O + O_2$

$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$

$SO_3 + H_2O \rightarrow H_2SO_4$

$2NaHCO_3 \rightarrow Na_2CO_3 + CO_2 + H_2O$

$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

$AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

Q7 CR1-LO3

Use the activity series (K Ca Na Mg Al Zn Fe Pb (H) Cu Ag Au). Which reaction will happen?

A. $Cu + ZnSO_4 \rightarrow CuSO_4 + Zn$

B. $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

C. $2Ag + 2HCl \rightarrow 2AgCl + H_2$

D. $Br_2 + 2KCl \rightarrow 2KBr + Cl_2$

Q8 CR1-LO3

Complete the double displacement used to make the barium meal for X-rays: $BaCl_2(aq) + Na_2SO_4(aq) \rightarrow$ _____(s) + 2NaCl(aq). Write the formula of the solid.

Q9 CR1-LO3

The antacid reaction $Al(OH)_3 + 3HCl \rightarrow AlCl_3 + 3H_2O$ is a double displacement reaction.

True / False

Q10 CR1-LO3

A double displacement reaction happens when one product leaves the solution. Which of these is **not** one of the signs?

A. A solid (precipitate) forms

B. Water forms

C. A gas bubbles out

D. All the products stay dissolved

Parts 4–5 · Redox

Q11 CR1-LO4

In $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$, which substance is oxidized?

A. Zn

B. Cu^{2+}

C. Zn^{2+}

D. Cu

Q12 CR1-LO4

In the pacemaker battery reaction $2Li + I_2 \rightarrow 2LiI$, the reducing agent is:

A. I_2

B. Li

C. LiI

D. none

Q13 CR1-LO4

Complete the half reaction: $Al \rightarrow Al^{3+} + __e^-$

_____

Q14 CR1-LO4

Is each change an oxidation or a reduction?

Options: oxidation · reduction

ethanol → acetaldehyde (loses 2 H) _____

pyruvic acid → lactic acid (gains 2 H) _____

$CH_4 \rightarrow CO_2$ _____

CO_2 → glucose (photosynthesis) _____

Q15 CR1-LO4

An antioxidant such as vitamin C works as a reducing agent.

True / False

Part 6 · The mole and molar mass

Q16 CR1-LO5

Calculate the molar mass of CO_2 (C = 12.01, O = 16.00).

_____

Q17 CR1-LO5

How many moles are in 36.04 g of water (18.02 g/mol)?

_____

Q18 CR1-LO5

How many moles of O atoms are in 2 mol of H_2SO_4?

A. 2

B. 4

C. 6

D. 8

Q19 CR1-LO5

How many molecules are in 0.500 mol of glucose?

A. 0.500

B. 3.011×10^{23}

C. 6.022×10^{23}

D. 1.204×10^{24}

Part 7 · Stoichiometry

Q20 CR1-LO6

Using $2H_2 + O_2 \rightarrow 2H_2O$, how many moles of O_2 are needed to make 5.0 mol of H_2O?

Q21 CR1-LO6

$CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$. What mass of CO_2 (44.01 g/mol) forms from 10.0 g of $CaCO_3$ (100.09 g/mol)?

Pharmacy scenario · An antacid tablet

Scenario

A milk of magnesia tablet contains **500 mg of magnesium hydroxide**. It neutralizes stomach acid: **$Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$**. (Mg = 24.31, O = 16.00, H = 1.01, Cl = 35.45)

Q22 CR1-LO5

Calculate the molar mass of $Mg(OH)_2$.

Q23 CR1-LO5

How many moles of $Mg(OH)_2$ are in the 500 mg tablet? (Give 3 significant figures.)

Q24 CR1-LO6

How many moles of HCl can the tablet neutralize?

Q25 CR1-LO6

What mass of HCl is that (HCl = 36.46 g/mol)? Explain why the answer is not simply 0.500 g.

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Periodic table · each box: atomic number, symbol, atomic mass, **electronegativity** · colours: metal, metalloid, nonmetal, noble gas

1A 1	2A 2	3	4	5	6	7	8	9	10	11	12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 **H** 1.01 2.1																	2 **He** 4.00 —
3 **Li** 6.94 1.0	4 **Be** 9.01 1.5											5 **B** 10.81 2.0	6 **C** 12.01 2.5	7 **N** 14.01 3.0	8 **O** 16.00 3.5	9 **F** 19.00 4.0	10 **Ne** 20.18 —
11 **Na** 22.99 0.9	12 **Mg** 24.31 1.2											13 **Al** 26.98 1.5	14 **Si** 28.09 1.8	15 **P** 30.97 2.1	16 **S** 32.07 2.5	17 **Cl** 35.45 3.0	18 **Ar** 39.95 —
19 **K** 39.10 0.8	20 **Ca** 40.08 1.0	21 **Sc** 44.96 1.3	22 **Ti** 47.87 1.5	23 **V** 50.94 1.6	24 **Cr** 52.00 1.6	25 **Mn** 54.94 1.5	26 **Fe** 55.84 1.8	27 **Co** 58.93 1.9	28 **Ni** 58.69 1.9	29 **Cu** 63.55 1.9	30 **Zn** 65.38 1.6	31 **Ga** 69.72 1.6	32 **Ge** 72.63 1.8	33 **As** 74.92 2.0	34 **Se** 78.97 2.4	35 **Br** 79.90 2.8	36 **Kr** 83.80 —
37 **Rb** 85.47 0.8	38 **Sr** 87.62 1.0	39 **Y** 88.91 1.2	40 **Zr** 91.22 1.4	41 **Nb** 92.91 1.6	42 **Mo** 95.95 1.8	43 **Tc** [98] 1.9	44 **Ru** 101.07 2.2	45 **Rh** 102.91 2.2	46 **Pd** 106.42 2.2	47 **Ag** 107.87 1.9	48 **Cd** 112.41 1.7	49 **In** 114.82 1.7	50 **Sn** 118.71 1.8	51 **Sb** 121.76 1.9	52 **Te** 127.60 2.1	53 **I** 126.90 2.5	54 **Xe** 131.29 —
55 **Cs** 132.91 0.7	56 **Ba** 137.33 0.9	57–71	72 **Hf** 178.49 1.3	73 **Ta** 180.95 1.5	74 **W** 183.84 1.7	75 **Re** 186.21 1.9	76 **Os** 190.23 2.2	77 **Ir** 192.22 2.2	78 **Pt** 195.08 2.2	79 **Au** 196.97 2.4	80 **Hg** 200.59 1.9	81 **Tl** 204.38 1.8	82 **Pb** 207.2 1.9	83 **Bi** 208.98 1.9	84 **Po** [209] 2.0	85 **At** [210] 2.2	86 **Rn** [222] —
87 **Fr** [223] 0.7	88 **Ra** [226] 0.9	89–103	104 **Rf** [267] —	105 **Db** [268] —	106 **Sg** [269] —	107 **Bh** [270] —	108 **Hs** [277] —	109 **Mt** [278] —	110 **Ds** [281] —	111 **Rg** [282] —	112 **Cn** [285] —	113 **Nh** [286] —	114 **Fl** [289] —	115 **Mc** [290] —	116 **Lv** [293] —	117 **Ts** [294] —	118 **Og** [294] —

	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Lanthanides	57 **La** 138.91 —	58 **Ce** 140.12 —	59 **Pr** 140.91 —	60 **Nd** 144.24 —	61 **Pm** [145] —	62 **Sm** 150.36 —	63 **Eu** 151.96 —	64 **Gd** 157.25 —	65 **Tb** 158.93 —	66 **Dy** 162.50 —	67 **Ho** 164.93 —	68 **Er** 167.26 —	69 **Tm** 168.93 —	70 **Yb** 173.05 —	71 **Lu** 174.97 —
Actinides	89 **Ac** [227] 1.1	90 **Th** 232.04 1.3	91 **Pa** 231.04 1.4	92 **U** 238.03 1.4	93 **Np** [237] —	94 **Pu** [244] —	95 **Am** [243] —	96 **Cm** [247] —	97 **Bk** [247] —	98 **Cf** [251] —	99 **Es** [252] —	100 **Fm** [257] —	101 **Md** [258] —	102 **No** [259] —	103 **Lr** [262] —

Atomic masses to 2 d.p. ([n] = mass number of the most stable isotope). Electronegativity: Pauling values as in Ball, *The Basics of GOB Chemistry*, Figure 4.4.3 (La–Lu 1.0–1.2; noble gases have no value).

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About 30 minutes. Questions follow the seven parts of the lecture and end with a pharmacy scenario.

Answer key

Parts 1–2 · Equations and ratios

Q1 CR1-LO1

Which is the balanced equation for iron rusting, $Fe + O_2 \rightarrow Fe_2O_3$?

A. $2Fe + O_2 \rightarrow Fe_2O_3$

B. $4Fe + 3O_2 \rightarrow 2Fe_2O_3$

C. $Fe + O_3 \rightarrow FeO_3$

D. $2Fe + 3O_2 \rightarrow Fe_2O_3$

4 Fe and 6 O on each side, in the lowest ratio.

Q2 CR1-LO1

Balance $N_2 + __H_2 \rightarrow 2NH_3$. What coefficient goes in front of H_2?

3

$2NH_3$ contains 6 H, so $3H_2$ is needed.

Q3 CR1-LO1

It is acceptable to change a subscript in a formula to balance an equation.

True / False **→ False**

Changing a subscript changes the substance. Only coefficients change.

Q4 CR1-LO1

12.0 g of magnesium burns completely in 8.0 g of oxygen to form magnesium oxide only. What mass of magnesium oxide forms?

20 g

Conservation of mass: 12.0 + 8.0 = 20.0 g.

Q5 CR1-LO2

Using $2C_2H_2 + 5O_2 \rightarrow 4CO_2 + 2H_2O$, how many O_2 molecules react with 10 C_2H_2 molecules?

25 molecules

10 × (5 O_2 / 2 C_2H_2) = 25.

Part 3 · Types of reaction

Q6 CR1-LO3

Classify each reaction.

$2H_2O_2 \rightarrow 2H_2O + O_2$ → decomposition; $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$ → combustion; $SO_3 + H_2O \rightarrow H_2SO_4$ → combination; $2NaHCO_3 \rightarrow Na_2CO_3 + CO_2 + H_2O$ → decomposition; $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$ → single displacement; $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$ → double displacement

One → many is decomposition; + O_2 giving oxides is combustion; many → one is combination; an element pushing another out is single displacement; two compounds swapping ions is double displacement.

Q7 CR1-LO3

Use the activity series (K Ca Na Mg Al Zn Fe Pb (H) Cu Ag Au). Which reaction will happen?

A. $Cu + ZnSO_4 \rightarrow CuSO_4 + Zn$

B. $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

C. $2Ag + 2HCl \rightarrow 2AgCl + H_2$

D. $Br_2 + 2KCl \rightarrow 2KBr + Cl_2$

Fe is above Cu, so Fe can push Cu out. Cu is below Zn, Ag is below H, and Br_2 is below Cl_2.

Q8 CR1-LO3

Complete the double displacement used to make the barium meal for X-rays: $BaCl_2(aq) + Na_2SO_4(aq) \rightarrow$ _____$(s) + 2NaCl(aq)$. Write the formula of the solid.

$BaSO_4$

Ba^{2+} pairs with SO_4^{2-} to give solid barium sulfate, $BaSO_4$.

Q9 CR1-LO3

The antacid reaction $Al(OH)_3 + 3HCl \rightarrow AlCl_3 + 3H_2O$ is a double displacement reaction.

True / False **→ True**

Two compounds swap ions, and water forms. This is a neutralization, a kind of double displacement.

Q10 CR1-LO3

A double displacement reaction happens when one product leaves the solution. Which of these is **not** one of the signs?

A. A solid (precipitate) forms

B. Water forms

C. A gas bubbles out

D. All the products stay dissolved

If every product stays dissolved, the ions just stay mixed and no reaction happens.

Parts 4–5 · Redox

Q11 CR1-LO4

In $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$, which substance is oxidized?

A. Zn

B. Cu^{2+}

C. Zn^{2+}

D. Cu

Zn loses two electrons ($Zn \rightarrow Zn^{2+} + 2e^-$).

Q12 CR1-LO4

In the pacemaker battery reaction $2Li + I_2 \rightarrow 2LiI$, the reducing agent is:

A. I_2

B. Li

C. LiI

D. none

Li is oxidized to Li^+, so it is the reducing agent.

Q13 CR1-LO4

Complete the half reaction: $Al \rightarrow Al^{3+} + __e^-$

3

Aluminium loses three electrons; the charge is 0 on each side.

Q14 CR1-LO4

Is each change an oxidation or a reduction?

ethanol → acetaldehyde (loses 2 H) → oxidation; pyruvic acid → lactic acid (gains 2 H) → reduction; $CH_4 \rightarrow CO_2$ → oxidation; CO_2 → glucose (photosynthesis) → reduction

Lose H or gain O = oxidation; gain H or lose O = reduction.

Q15 CR1-LO4

An antioxidant such as vitamin C works as a reducing agent.

True / False **→ True**

It is oxidized itself, protecting other molecules from oxidation.

Part 6 · The mole and molar mass

Q16 CR1-LO5

Calculate the molar mass of CO_2 (C = 12.01, O = 16.00).

44.01 g/mol

12.01 + 2 × 16.00 = 44.01 g/mol.

Q17 CR1-LO5

How many moles are in 36.04 g of water (18.02 g/mol)?

2 mol

36.04 g ÷ 18.02 g/mol = 2.000 mol.

Q18 CR1-LO5

How many moles of O atoms are in 2 mol of H_2SO_4?

A. 2

B. 4

C. 6

D. 8

Each H_2SO_4 has 4 O, so 2 × 4 = 8 mol O.

Q19 CR1-LO5

How many molecules are in 0.500 mol of glucose?

A. 0.500

B. 3.011×10^{23}

C. 6.022×10^{23}

D. 1.204×10^{24}

$0.500 \times 6.022 \times 10^{23} = 3.011 \times 10^{23}$.

Part 7 · Stoichiometry

Q20 CR1-LO6

Using $2H_2 + O_2 \rightarrow 2H_2O$, how many moles of O_2 are needed to make 5.0 mol of H_2O?

2.5 mol

5.0 × (1 mol O_2 / 2 mol H_2O) = 2.5 mol.

Q21 CR1-LO6

$CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$. What mass of CO_2 (44.01 g/mol) forms from 10.0 g of $CaCO_3$ (100.09 g/mol)?

4.4 g

10.0 ÷ 100.09 = 0.0999 mol $CaCO_3$ → 0.0999 mol CO_2 (1 : 1) × 44.01 = 4.40 g.

Pharmacy scenario · An antacid tablet

Q22 CR1-LO5

Calculate the molar mass of $Mg(OH)_2$.

58.33 g/mol

24.31 + 2 × (16.00 + 1.01) = 58.33 g/mol.

Q23 CR1-LO5

How many moles of $Mg(OH)_2$ are in the 500 mg tablet? (Give 3 significant figures.)

0.00857 mol

0.500 g ÷ 58.33 g/mol = 0.00857 mol.

Q24 CR1-LO6

How many moles of HCl can the tablet neutralize?

0.0171 mol

0.00857 mol × (2 mol HCl / 1 mol $Mg(OH)_2$) = 0.0171 mol.

Q25 CR1-LO6

What mass of HCl is that (HCl = 36.46 g/mol)? Explain why the answer is not simply 0.500 g.

0.0171 mol × 36.46 g/mol = **0.625 g HCl**. Grams cannot be compared directly, because $Mg(OH)_2$ and HCl have different molar masses and react in a 1 : 2 **mole** ratio. Always go grams → moles → mole ratio → moles → grams.

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